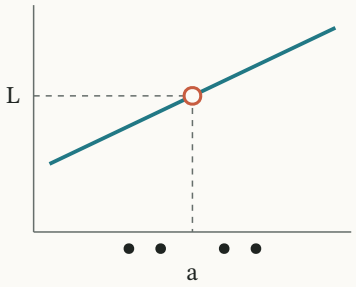
Limits CHEAT SHEET

THE IDEA OF A LIMIT



where f(x) is heading as x gets close to a, never at a itself

$$\lim_{x\to a} f(x) = L$$

$\lim_{x\to a^-}$ left-hand $\lim_{x\to a^+}$ right-hand

the limit exists $\Leftrightarrow$ left = right

LIMIT LAWS

$\lim (f \pm g) = \lim f \pm \lim g$

$\lim (k\,f) = k \lim f$

$\lim (f\,g) = \lim f \cdot \lim g$

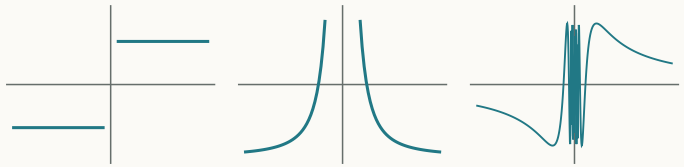
$\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ if $\lim g \neq 0$

$\lim f^n = (\lim f)^n$ $\quad \lim \sqrt{f} = \sqrt{(\lim f)}$

polynomial p: $\lim_{x\to a} p(x) = p(a)$

rational p/q: substitute when $q(a) \neq 0$

WHEN THERE IS NO LIMIT



sides differ · blows up · never settles

ZERO OVER ZERO

1 try the number first

2 0/0 is not an answer: look closer

factorise: $\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3} (x+3) = 6$

rationalise: $\lim_{x\to 0} \frac{\sqrt{(x+4)}-2}{x} = \lim_{x\to 0} \frac{1}{\sqrt{(x+4)}+2} = \frac{1}{4}$

$\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}$ $\quad$ e.g. $\lim_{x\to 2} \frac{x^5-32}{x-2} = 80$

STANDARD LIMITS

$\lim_{x\to 0} \frac{\sin x}{x}$	$= 1$
$\lim_{x\to 0} \frac{\tan x}{x}$	$= 1$
$\lim_{x\to 0} \frac{1-\cos x}{x}$	$= 0$
$\lim_{x\to 0} \frac{1-\cos x}{x^2}$	$= 1/2$
$\lim_{n\to\infty} (1+\frac{1}{n})^n$	$= e \approx 2.718$
$\lim_{x\to 0} (1+x)^{1/x}$	$= e$
$\lim_{x\to 0} \frac{e^x-1}{x}$	$= 1$
$\lim_{x\to 0} \frac{a^x-1}{x}$	$= \ln a$
$\lim_{x\to 0} \frac{\ln(1+x)}{x}$	$= 1$

x in radians · match the inside: sin 4x / 4x → 1

LIMITS INVOLVING INFINITY

$\lim_{x\to\infty} \frac{1}{x} = 0$ $\quad \lim_{x\to 0^+} \frac{1}{x} = \infty$ $\quad \lim_{x\to 0^-} \frac{1}{x} = -\infty$

$\lim_{x\to\infty} \frac{2x^2+3x}{5x^2-1} = \frac{2}{5}$ (divide by x^2)

$\lim_{x\to-\infty} \frac{3x}{\sqrt{(x^2+1)}} = -3$ $\quad (\sqrt{(x^2)} = |x|)$

rational, x → ±∞: compare the highest powers

- top lower → 0
- equal → ratio of leading numbers
- top higher → no finite limit

vertical asymptote x = a: a one-sided limit is ±∞
horizontal asymptote y = L: the limit at ±∞ is L

INDETERMINATE FORMS

0/0	$\lim_{x\to 3} \frac{x^2-9}{x-3} = 6$
∞/∞	$\lim_{x\to\infty} \frac{2x^2+3x}{5x^2-1} = \frac{2}{5}$
∞ − ∞	$\lim_{x\to\infty} (\sqrt{(x^2+4x)} - x) = 2$
1^∞	$\lim_{x\to 0} (1+3x)^{1/x} = e^3 \approx 20.09$

none of these is an answer: each needs a closer look

MATCH THE INSIDE

$\lim_{x\to 0} \frac{\sin 4x}{x} = 4 \cdot \frac{\sin 4x}{4x} \to 4$

$\lim_{x\to 0} \frac{\tan 3x}{\sin 2x} = \frac{3}{2}$

$\lim_{x\to 0} \frac{1-\cos 2x}{x^2} = 2$

$\lim_{x\to 0} \frac{e^{5x}-1}{x} = 5$

$\lim_{x\to 0} \frac{\ln(1+3x)}{\sin x} = 3$

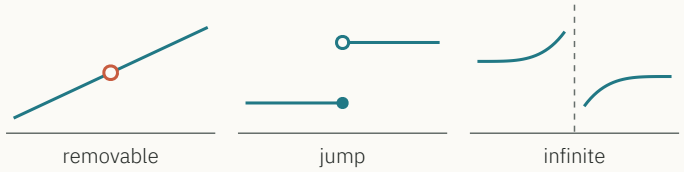
$\lim_{n\to\infty} (1-\frac{1}{n})^n = \frac{1}{e} \approx 0.368$

Limits CHEAT SHEET

CONTINUITY

continuous at a: $\lim_{x\to a} f(x) = f(a)$

f(a) exists · the limit exists · they are equal



polynomials, sin, cos, e^x: everywhere · ln x: x > 0

MAKING IT CONTINUOUS

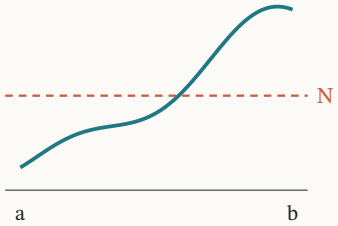
$kx + 1$ ($x \le 2$), $x^2 - 1$ ($x > 2$): $2k + 1 = 3$, $k = 1$

$\frac{\sin x}{x}$ ($x \neq 0$), k ($x = 0$): $k = 1$

$\frac{x^2 - 4}{x - 2}$: define $f(2) = 4$

removable: the limit exists; the value is missing or wrong
jump: the one-sided limits differ · infinite: one is ±∞

INTERMEDIATE VALUE THEOREM

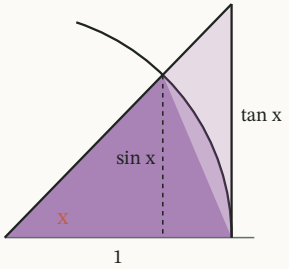


f continuous on [a, b],
N between f(a) and f(b)
⇒ f(c) = N for some c
in (a, b): at least one

$x^3 + x - 1$: f(0) = −1, f(1) = 1, so a root in (0, 1), ≈ 0.682

THE SQUEEZE THEOREM

$g \le f \le h$ near a, $\lim g = \lim h = L \Rightarrow \lim f = L$

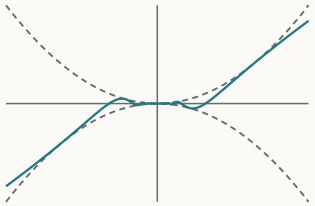


$\frac{1}{2}\sin x < \frac{1}{2}x < \frac{1}{2}\tan x$

$\cos x < \frac{\sin x}{x} < 1$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

THE SQUEEZE IN ACTION



$-x^2 \le x^2 \sin(1/x) \le x^2$

both sides → 0

so $\lim_{x\to 0} x^2 \sin(1/x) = 0$

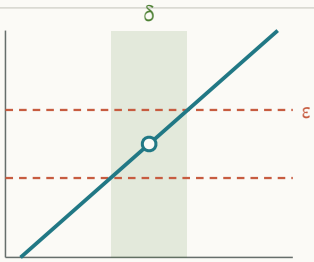
WHY RADIANS

$\sin 1° \div 1 \approx 0.01745 = \pi/180$

small x, in radians: $\sin x \approx x$ $\quad \sin 0.05 \approx 0.04998$

in radians, the arc on a circle of radius 1 is the angle itself

THE PRECISE DEFINITION



for every ε > 0
there is a δ > 0:
$0 < |x - a| < \delta$
$\Rightarrow |f(x) - L| < \varepsilon$

2x + 1 → 7 at 3: δ = ε/2

x → ∞: for every ε there is an N with x > N ⇒ |f(x) − L| < ε

FOUR EPSILON-DELTA RESULTS

$2x + 1 \to 7$ as $x \to 3$: $\delta = \varepsilon/2$

$x^2 \to 4$ as $x \to 2$: $\delta = \min(1, \varepsilon/5)$

$\frac{|x|}{x}$ at 0: $\varepsilon = \frac{1}{2}$ fails every δ (values −1 and 1)

$1/x \to 0$ as $x \to \infty$: $N = 1/\varepsilon$

THE SLOPE OF A CURVE

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

the chord's slope, as the step h shrinks

x^2 at 3: $\frac{(3+h)^2 - 9}{h} = 6 + h \to 6$

$\sin x / x$, $(e^x - 1)/x$ and $(x^n - a^n)/(x - a)$ are slopes